

EXAMEN SM 2023 Subiectul (Randul 1)

Probabilitati + Combinatorica

1. We have 20 letter out of which 6 are registered. In how many ways can a mailman split the letters: a) in 2 piles b) in 4 equal piles c) in 10 mailboxes...
2. a) how many 5 digit numbers exist?
b) how many number between 1 and 10^6-1 have: b1) the sum of digits = 9
b2) the sum of digits = 19
b3) the sum of digits ≤ 9
3. 2 hunters shoot a target at the same time, with probability to hit 0.6 and 0.4 respectively. What is the probability that:
a) Both hunters hit the target
b) The target is hit by at least one hunter
c) The second target hits the target given that the target was hit
d) ... nu mai tin minte. Aici atentie ca loviturile nu formeaza CSE, probema asta o face mereu la seminar sa fiti atenti la ea, o si da des.
4. In an urn we have 2 white balls and 1 black ball. We extract one ball from the urn. If it is white, the experiment ends, if it is black we add it back to the urn along with another white ball. Let X be the nr of extractions until the apparition of a white ball. Calculate: a) $P(X=2)$
b) $P(X \geq 4)$ c) Write the distribution of X and prove that it is correctly defined d) $E(X) = ?$.
5. Write the idee expressed by the law of large numbers for the random variable X in the previous problem. (Scriteti LLN corect ca e important si da punctaj maxim pe el si daca nu ai stiut calcula $E(x)$ precedent)

Grafuri

1. Aici erau grafurile pe table deci nu le pot da dar s-a cerut MAX FLOW/MIN CUT.
2. Alt graf pe tabla, se cerea a) write a matching with $\text{card}(M) > 2$ that is not maximum (and explain why it is not maximum) for the given graph. b) Apply Dijkstra's algorithm starting from vertex h and at each step list all the candidate edges and justify your decisions. Color the output on the graph. What do the labels on the vertices represent and what problem is solved by this algorithm?
3. De demonstrat o teorema cu matching si augmenting path nu am scris nimic de aia nu o stiu.
4. How can we establish in the quickest way the WEAK connectivity of a directed graph having at our disposal a BOOLEAN ADJACENCT MATRIX FOR THE DIGRAPH (toata problema asta o fost discutata ideea la un seminar intreg).
5. Inca o problema de matching, again n-am invatat matching nu mi-am pierdut timpul.

SUCCES